STE 73822-1

INSTRUCTION MANUAL

INDUSTRIAL ROBOT SR SERIES

TRANSPORTATION AND INSTALLATION MANUAL

Notice

1. Make sure that this Instruction Manual is delivered to the final user of the Industrial Robot.

2. Please read this manual before installing or operating the Industrial Robot, and keep the manual nearby for further reference during use of the robot.

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Shibaura Machine

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PREFACE

This Manual describes how to unpack and install the SR series large-sized SCARA robot and controller. Specifically, it describes how to remove the equipment from the boxes, how to position the equipment, how to connect wiring and air piping, and how to attach tools. Please look through this Manual before opening any boxes.

Before beginning work with this Manual, please read the Introductory Manual and Safety Manual so that you may learn the names and functions of the various components and the safety measures.

This Manual is divided into four chapters:

Chapter 1 Transportation

This chapter describes how to remove the robot and controller from their boxes and how to transport them to the installation site. This chapter also discusses how to temporarily store the equipment after opening the boxes.

Chapter 2 Installation

This chapter discusses the equipment installation environment, space requirements, and how to install the equipment.

Chapter 3 System Connections

This chapter describes how to connect the robot, controller and peripheral equipment.

Chapter 4 Tool Interlace

This chapter describes how to connect the tool to the robot arm and how to connect pipes and wires to the tool. This chapter also discusses maximum permissible loads for the tool.

Note) The SR series large-sized SCARA robot is a column type robot whose arm length is 1200 mm or over.

Example: 1504HZ

PRECAUTIONS FOR SAFETY

Important information is noted on the robot and controller, in the instruction manual to prevent injury to the user and persons near by, prevent damage to assets and to ensure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 Danger	This means that "incorrect handling will lead to fatalities or major injuries".
 Caution	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)".

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long term treatment.
*2) Physical damage refers to major fires due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited(must not be done). The details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory(must be done). The details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. The details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

Caution

- Always read the separate Instruction Manual "Safety Section" before starting actual work to ensure safe work from the robot installation to operation.

[Installation and transportation]

Always observe the following items to safely use the robot.

Danger	
 Prohibited	・ Do not install or operate if any parts are damaged or missing. Doing so could lead to electric shocks, fires or faults. ・ Do not install the robot where it may be subject to fluids such as water. Doing so could lead to electric shocks, fires or faults. ・ Do not place the robot near combustible matters. Doing so could lead to fires if the matter ignites due to a fault, etc.
 Mandatory	・ Always secure the robot with the exclusive carrier base before transporting it. Failure to do so could lead to injuries if the arm moves when the robot is suspended. ・ Wire the robot after installation. Wiring the robot before installation could lead to electric shocks or injuries. ・ Always use the power voltage and power capacity designated by SHIBAURA MACHINE. Failure to do so could lead to device faults or fires. ・ Always use the designated power cable. Using a cable other than that designated could lead to fires or faults.
 Always ground	・ Securely connect the grounding cable. Failure to do so could lead to electric shocks or fires if a fault or fault current occurs. Noise could lead to malfunction. Also, it could cause mis-operation by noise.

Caution	
 Prohibited	・ Never lift arm, etc. Doing so will apply an excessive force on the robot's mechanism section and could lead to faults. ・ For the controller, secure the ample space for air vent. Heating of controller could lead to malfunction.
 Mandatory	・ When lifting the robot, lift it up slowly. The robot will tilt slightly, so lifting it up suddenly could be hazardous. ・ When storing the robot, securely fix it on the base. The robot will be unstable if just set down, and it could tilt over.

TABLE OF CONTENTS

CHAPTER 1 TRANSPORTATION

1.1 UNPACKING **1-1**
1.2 TRANSPORTATION **1-2**
1.2.1 Mass and Dimensions 1-2
1.2.2 Transporting the Robot 1-2
1.2.3 Transporting the Controller 1-4
1.3 STORAGE **1-5**
1.3.1 Storage Precautions for the Robot 1-5
1.3.2 Storage Precautions for the Controller 1-5

CHAPTER 2 INSTALLATION

2.1 INSTALLATION ENVIRONMENT **2-1**
2.2 ROBOT INSTALLATION **2-1**
2.2.1 External Dimensions and Working Space 2-1
2.2.2 Coordinate System 2-3
2.2.3 Installing the Robot 2-4
2.3 INSTALLING THE CONTROLLER **2-5**
2.3.1 External Dimensions 2-5
2.3.2 Keeping Space for Ventilation and Maintenance 2-5
2.4 PRECAUTIONS FOR HANDLING THE TEACH PENDANT **2-6**
2.5 SAFETY MEASURES **2-6**

CHAPTER 3 SYSTEM CONNECTIONS

3.1 CABLE WIRING **3-1**
3.1.1 Connector Arrangement on the Controller 3-1
3.1.2 Connecting the Power Cable “CNPW” 3-2
3.1.3 Connecting the Motor Cable 3-2
3.1.4 Connecting the Encoder Cable “CN3” 3-3
3.1.5 Connecting the Robot Control Signal Cable “CN4” 3-4
3.1.6 Inserting and Removing Cables 3-4
3.1.7 List of Connector Terminal Arrangement Examples 3-5
3.2 CONTROLLER CONNECTOR SIGNALS **3-6**
3.2.1 Connector Signal Connection Diagrams 3-6

3.2.2 Jumpers for Safety Related Signals · · · · · · · · · · · · · 3-6

CHAPTER 4 TOOL INTERFACE

4.1 MOUNTING THE TOOL · 4-1
4.2 TOOL WIRING AND PIPING · · · · · · · · · · · · · · · · · · · 4-2
4.2.1 Tool Wiring · 4-2
4.2.2 Tool Air Piping · 4-6
4.2.3 Mounting Solenoid Valves, etc. on Arm · · · · · · · · · · · 4-7
4.3 PERMISSIBLE LOAD CONDITIONS AND SETTING OF THE PROGRAM · 4-8
4.3.1 Permissible Load Conditions · · · · · · · · · · · · · · · · · 4-8
4.3.2 Load Conditions and Setting of the Program · · · · · · · · 4-9
4.3.3 Setting Robot Acceleration/Deceleration Time for Load Conditions · 4-12

APPENDIX A Calculating Control Power Capacity · · · · · · · · · · 5-1
APPENDIX B Restoring Zero Position Data (Restoration by Visual Check of Zero-Point Marks) · · · · · · · · · · · · · · · · · · 5-2
APPENDIX C Restoring Zero Position Data (Restoration by Zero Point Setting Tool) · 5-7

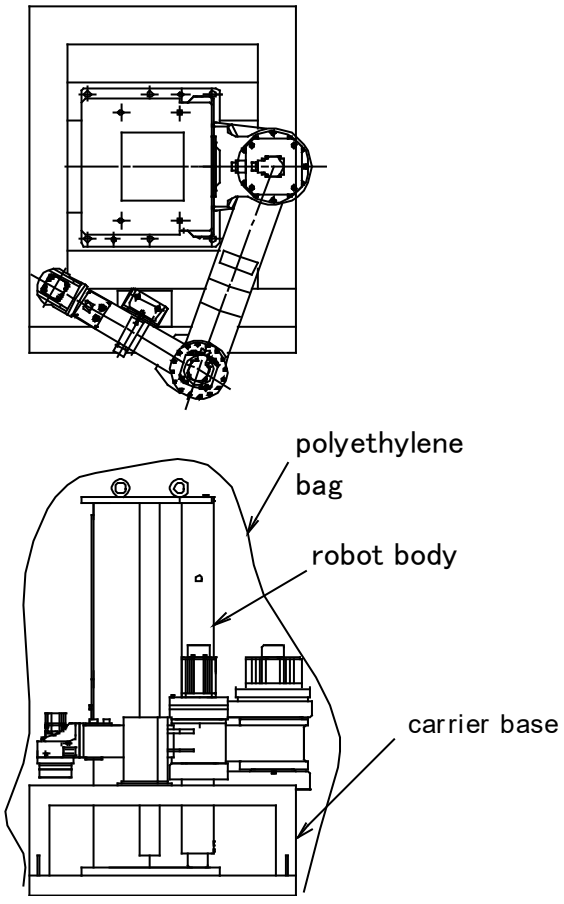
CHAPTER 1

TRANSPORTATION

1.1 UNPACKING

The robot and controller are shipped separately. The robot packaging state is shown in Fig. 1.1. The teach pendant, cables and other accessories such as the instruction manual are packaged separately.

Open the packages in a location easily accessible to where the equipment is to be installed. Take care not to damage the robot or controller. After opening the packages, make sure that all the accessories are present and that nothing has been damaged in transit.



Danger

- If damage is discovered of the parts such as the accessories are missing when the robot is delivered to the user, do not install or operate the unit.
  Doing so could lead to faults, so always contact SHIBAURA MACHINE.
- Dispose of the wooden frame, corrugated cardboard, polyethylene shipping bag and cushion material according to your standards.
- Return the carrier base after installation.

Fig. 1.1 Robot packaging state

1.2 TRANSPORTATION

Move the robot and controller very carefully. Make sure that no excessive impact (9.8m/s^2 or more) or vibration is applied on the equipment. If the equipment is to be subject to vibration over a long period, be sure to tighten all the nuts and screws.
If the equipment is to be moved to a location some distance from where it was unpacked, secure the arm and put the equipment back into the wooden frames and corrugated cardboards and reposition the cushions as they were.

1.2.1 Mass and Dimensions

Mass of the robot and controller is shown in Table 1.1 and outer dimensions of the packaging state are shown in Fig. 1.2.

Table 1.1

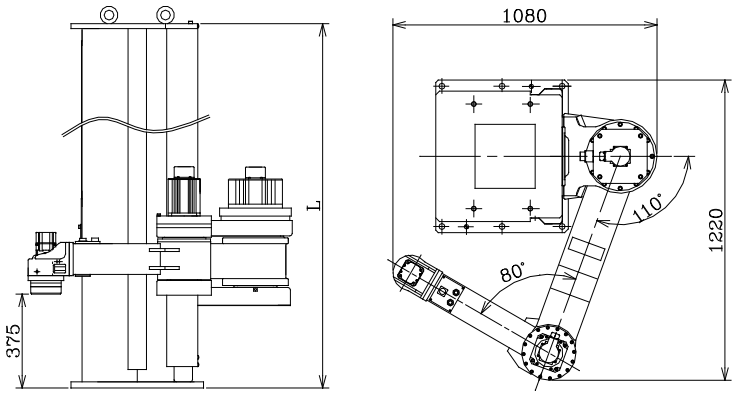
Type	3(Z)axis Stroke(mm)	Mass	Outer dimensions (during transport)
Robot SR–1504HZ	600	600 kg	See Fig. 1.2.
	1000	650 kg	
	1300	690 kg	
	1500	710 kg	
Controller TS2000L	90 kg	90 kg	See Fig. 1.3.

1.2.2 Transporting the Robot

As a general rule, transport the robot in the position shown in Fig. 1.2. The robot is shipped with the arm folded with a hold-down fixture, the primary side of Axes 1, 2 and 3 motor drive terminal blocks jumpered and the dynamic brake effected. (The shipment is made under this condition. After unpacking, transport the equipment as they are.)

Danger

- Always secure the arm with the dynamic brake before transporting the robot.
  Failure to do so could cause a hazardous situation as the arm will move when the robot is suspended.



3(Z)axis stroke(mm)	L(mm)
600	1480
1000	1880
1300	2180
1500	2380

Fig. 1.2 SR1504HZ robot packaging state

Lift the robot by crane. As shown in Fig. 1.3, attach wires to the four (4) M20 eyebolts on the column top. Then lift the robot carefully.

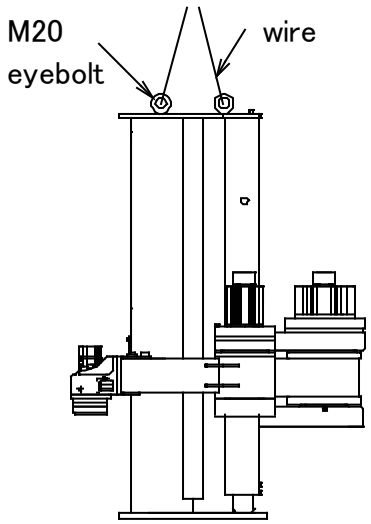


Fig. 1.3 Moving the robot by crane

Caution

- Use a wire with a limit load that can sufficiently withstand the robot mass.
- The robot will tilt somewhat when lifted, so slowly lift it up.
- When raising and lowering the robot, be careful not to apply any excessive impact (9.8m/s^2 or more) to the robot. Always treat it carefully.

1.2.3 Transporting the Controller

Disconnect all cables and teach pendant before transporting the controller.
As shown in Fig. 1.4, attach wires to the eyebolts on the top surface. Then lift the controller carefully.

Danger

- When placing the controller on the floor, etc., make sure not to catch your hands or feet.

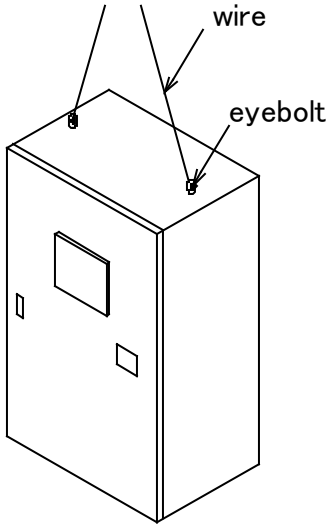


Fig. 1.4 Transporting the controller

1.3 STORAGE

We do not recommend storing the robot and controller for long periods of time after taking them out of their packages. However, when unavoidable, please observe the following precautions for storage.

1.3.1 Storage Precautions for the Robot

Caution

- Secure the robot completely in such a manner that it will not fall over. The robot is not stable when set on the floor directly and will fall over.
- Keep the robot out of direct sunlight. The timing belts may deteriorate.
- Seal the robot in a vinyl bag to prevent rusting and to keep out dust. Put a desiccant in the bag to absorb moisture. The column bottom plate, the ball screw and the linear guides are all susceptible to rusting. Therefore, coat the ball screw and linear guides with grease and the other surfaces with a rust preventative compound.
- Grease up the robot before use. (See the Maintenance Manual.)

1.3.2 Storage Precautions for the Controller

Caution

- Keep the controller out of direct sunlight.
  Direct sunlight can heat up the interior of the controller to temperatures.

CHAPTER 2

INSTALLATION

2.1 INSTALLATION ENVIRONMENT

Table 2.1 shows the environmental conditions for the location in which the Robot and TS2000L Controller are to be installed.

Table 2.1 Environmental Conditions for the Robot and Controller

Item	Specifications
Temperature	In operation : 0 to 40 ℃ In storage : –10 to 50 ℃
Humidity	20 to 90 % (Non-condensing) Do not install the robot where it may be subject to fluids such as water.
Altitude	1000 m or less
Vibration	In operation : Within 0.98m/s^2 In transport : Within 9.8m/s^2
Dust	There is to be no inductive dust. Consult with SHIBAURA MACHINE first, should you wish to use the robot and controller in an especially dusty environment.
Gas	There is to be no corrosive or combustible gas.
Sunlight	The robot must not be subject to direct sunlight.
Electromagnetic noise	There are to be no devices in the area which produce an excessive amount of electromechanical noise.
Field	There are to be no devices in the area that generate a strong field.

Danger

- Do not place the robot near combustible matters.
  Doing so could lead to fires if the matter ignites due to a fault, etc.

2.2 ROBOT INSTALLATION

Before actually installing the robot, you must plan a layout and take into consideration such matters as the working space, coordinate system and space for maintenance.

2.2.1 External Dimensions and Working Space

An external view and working space of the robot are shown in Fig. 2.1.
Each axis can operate within the working space. To keep the robot from moving out of the working space by mis-operation, the robot is equipped with mechanical stoppers outside the moving range (excluding the Axis 4). Additionally, soft limits are provided which may be set as desired by the user.

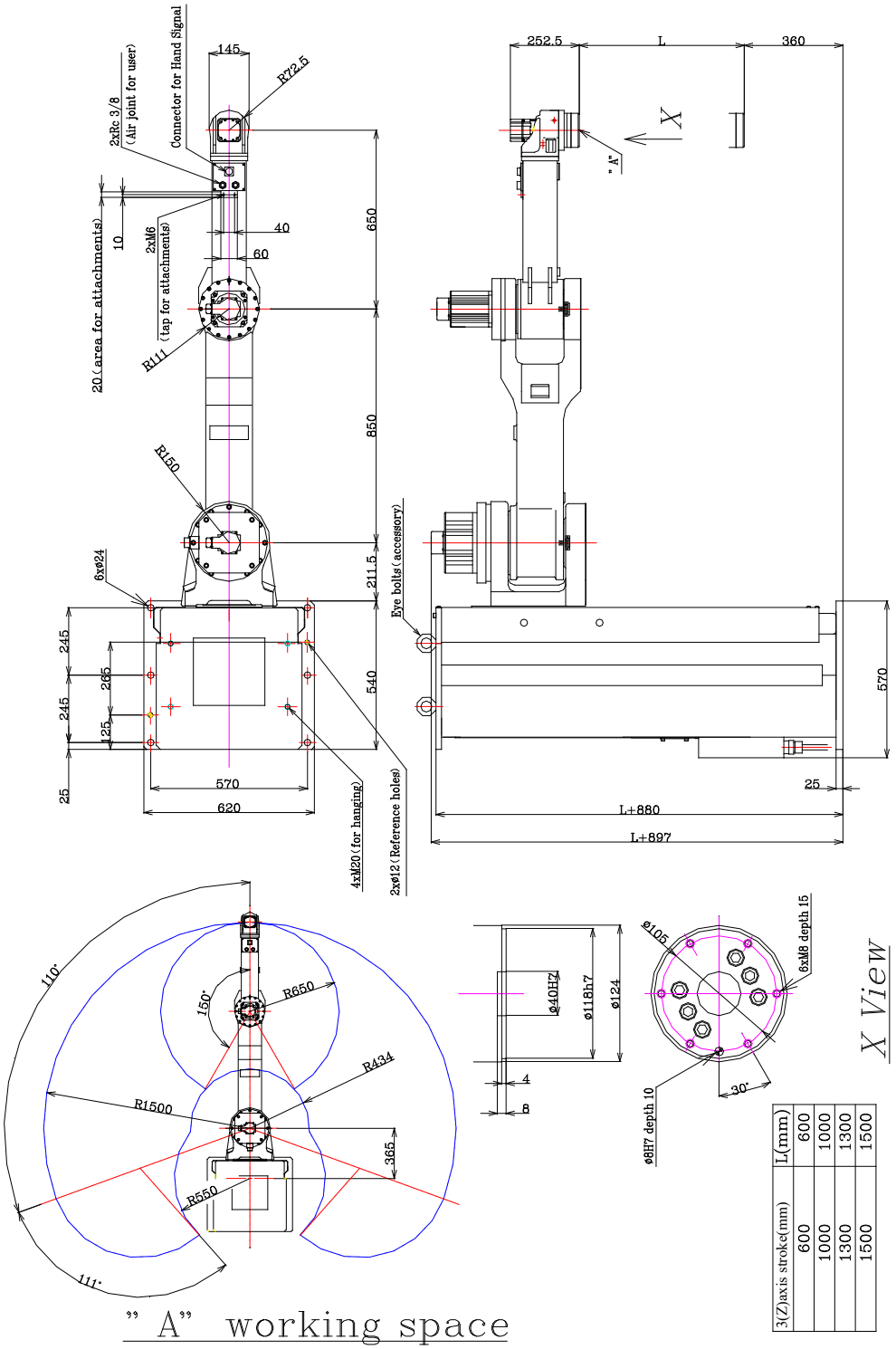


Fig. 2.1 SR–1504HZ external view and working space

2.2.2 Coordinate System

The robot's joint angle zero point (0° or 0 mm position) has been calibrated before shipment. The base coordinate system is determined according to this calibration. Fig. 2.2 shows the base coordinate system and the zero position of each axis joint angle.

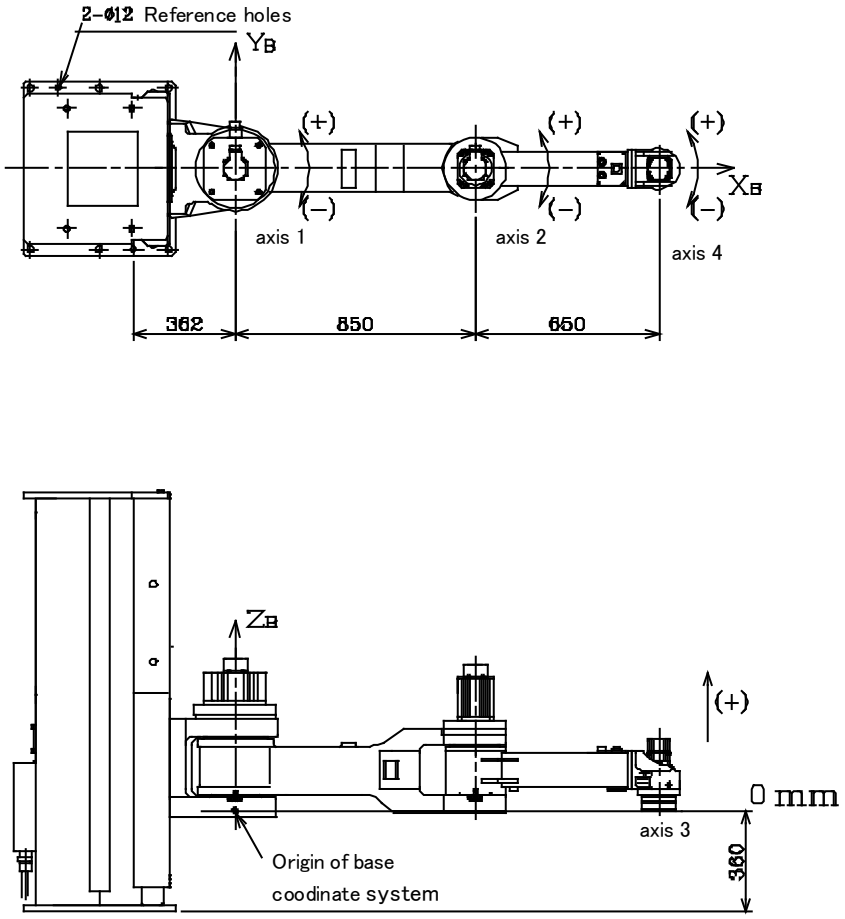


Fig. 2.2 Base coordinate system and joint angle zero point

2.2.3 Installing the Robot

The robot is secured, using the mounting holes in the base.
The robot installation method is shown in Fig. 2.3. A reference plane is provided on the base section, which is marked with "xxx". The locate holes (ø12 × 2) are also provided. To align the robot's base coordinate system position or when the robot must be replaced, use them. The reference plane has been coated with anticorrosive agents before shipment, so wipe the agent off with thinner before using.

Caution

- The robot will suddenly accelerate and decelerate during operation, so when installing it on a frame, make sure that the frame has sufficient strength and rigidity.
  If the robot is installed on a frame that does not have sufficient rigidity, the vibration will occur while the robot is operating, and could lead to faults.
  When installing the robot on the floor, fasten the robot down securely with foundation bolts.
- Install the robot on a level place.
  Failure to do so could lead to a drop in performance or faults.

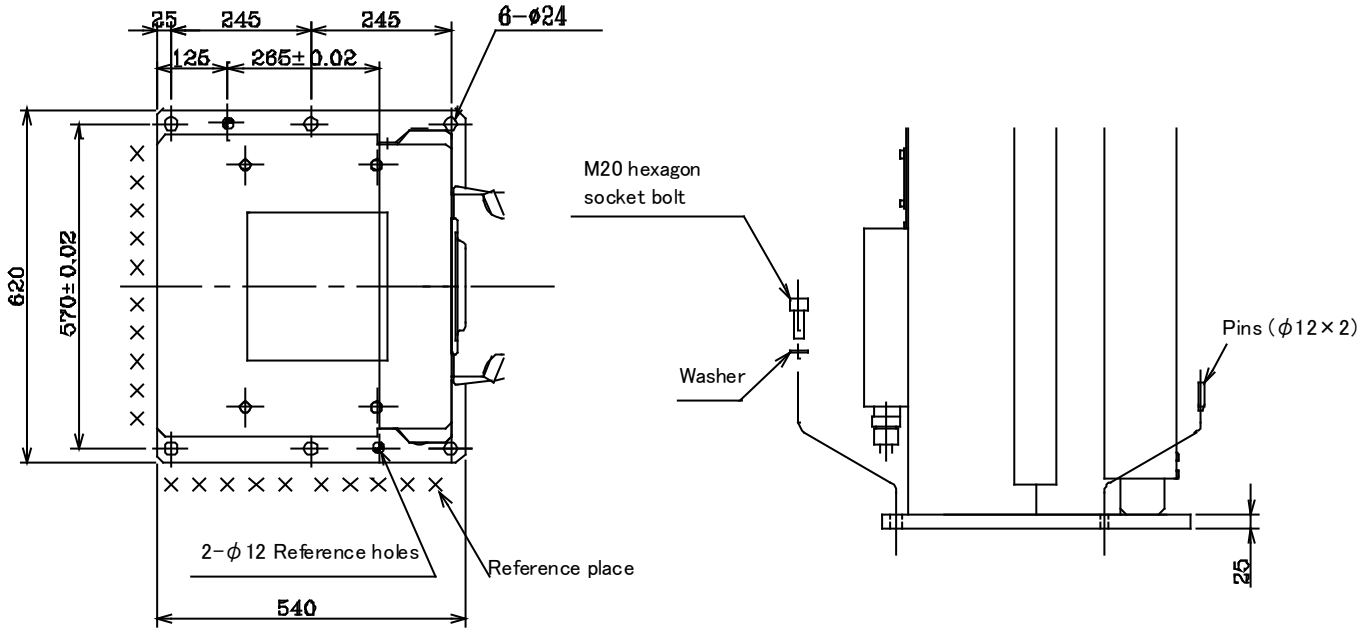


Fig. 2.3 Installing the robot

2.3 INSTALLING THE CONTROLLER

2.3.1 External Dimensions

An external view of the controller is shown in Fig 2.4.

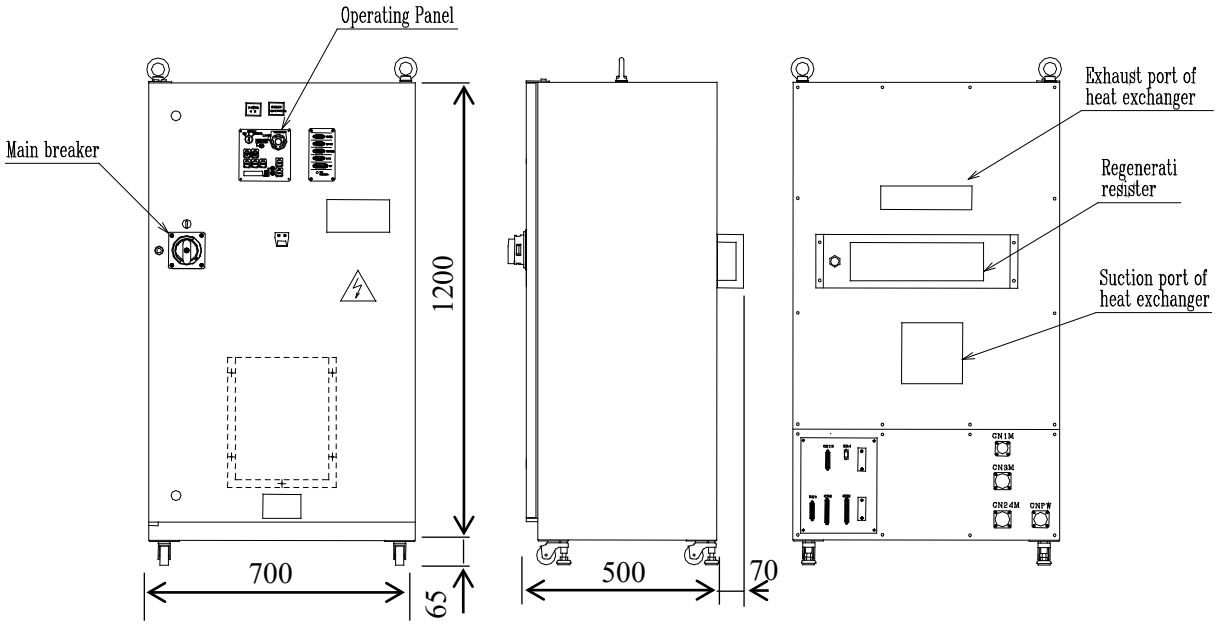


Fig. 2.4 External view of TS2000L controller

2.3.2 Keeping Space for Ventilation and Maintenance

Cooling in the controller is performed by means of the heat exchanger attached to the rear side. If the suction port and exhaust port are blocked, cooling ability will deteriorate to rise the temperature inside the controller, which results in a thermal trouble. Keep a clearance of more than 150 mm for the ventilation space at the rear side of the controller so that the controller interior can be cooled sufficiently.
For maintenance of the controller, open the front door.
When installing the controller, due care should be taken to keep a door space so that maintenance of the controller can be executed smoothly.

2.4 PRECAUTIONS FOR HANDLING THE TEACH PENDANT

Be careful of the following when handling the teach pendant.

Caution

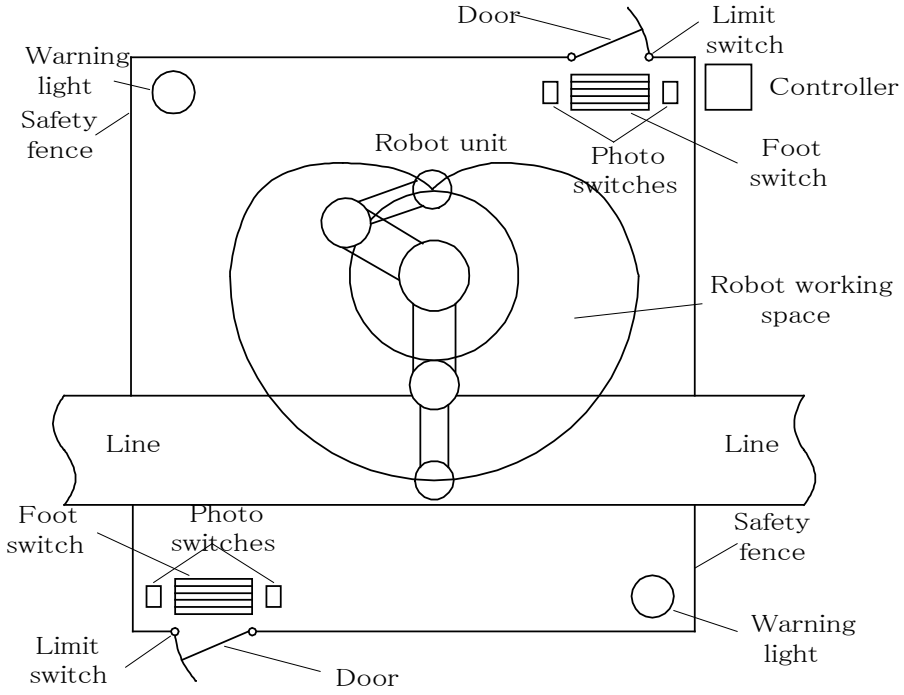
- Do not drop the teach pendant or hit it against anything.
- Do not pull on the cable protruding from the teach pendant.
- Do not press the switches on the teach pendant with anything sharp (like the tip of a knife, pencil, ball-point pen, etc.).
- Do not place or use the teach pendant near open flames.
- Do not leave the teach pendant in direct sunlight for a long period of time.

2.5 SAFETY MEASURES

a) When installing the robot, provide sufficient space to carry out the work safely.

b) Clarify the hazard zone, and provide a safety fence so that other persons cannot enter the zone easily. The hazard zone is the zone near the robot's working space where a hazardous state could occur if a person enters.

c) Provide a limit switch, photo switch or foot switch, etc., at the entrance of the safety fence to provide an emergency stop function that will stop the robot if a person enters the hazard zone. The emergency stop function should be an electrically independent b contact (closed in normal operation) with compulsive opening function and must not be automatically recovered.



d) The controller must be installed at a place outside the hazard zone where the operator can view the robot movement.

CHAPTER 3

SYSTEM CONNECTIONS

3.1 CABLE WIRING

This section describes the various types of cables and connectors and explains how these are to be hooked up.

3.1.1 Connector Arrangement on the Controller

The following cables are connected to the TS2000L robot controller:

[1] Power cable
[2] Motor cable
[3] Encoder cable
[4] Robot control signal cable
[5] External input/ output signal cable

They are connected to the back side of the controller as shown in Fig. 3.1.

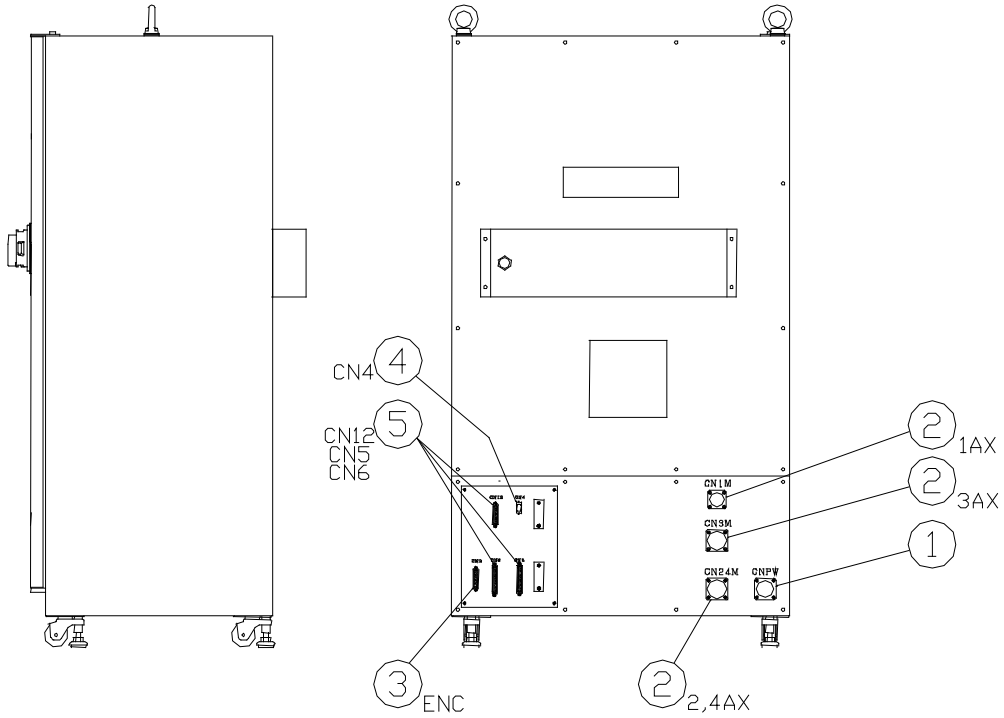


Fig. 3.1 Connector arrangement on the controller side

In this section, we explain how to connect cables [1] to [4]. For information on how to connect cables [5] to [9], refer to the Interface Manual.

3.1.2 Connecting the Power Cable "CNPW"

The power cable is used to supply the main AC power to the controller.
Fig. 3.1 [1]
The power supply specifications are shown in Table 3.1 below.

Table 3.1 Power supply specifications

Item	Specifications
Power supply	3-phase AC 180～250 V , 50/60Hz±1Hz
Power supply capacity	8 kVA
Instantaneous power failure	Within 40msec
Grounding	Class D dedicated grounding

The cable is not an accessory, which should be manufactured and provided by the customer.

CNPW CONNECTOR	Type: JL04V-6A22-22SE-EB	Maker:	Japan Aviation Electronics Industry
Clamp terminal	Type: JL04-2022CK(14)	Maker:	Japan Aviation Electronics Industry
Wire	5.5 mm^2		

Danger

- Always use the designated wire.
  Failure to do so could lead to fires or faults.
- When connecting the terminal block and wires, make sure not to mistake the arrangement of terminals and wires.

3.1.3 Connecting the Motor Cable ([3] of Fig. 3.1) (Cable included)

The motor cable connects the controller and robot and supplies the power required to rotate the motor from the controller's servo driver to each robot motor.
Axis 1 to 4 power is connected controller back side by connectors.

3.1.4 Connecting the Encoder Cable "CN3" ([3] of Fig. 3.1) (Cable included)

The encoder cable is a signal line used to transmit data from the encoders (devices used to detect the revolution (and thereby the position) of each shaft) to the controller.
The encoder cable runs from [3] of Fig. 3.1 (CN3) to [2] of Fig. 3.2.

3.1.5 Connecting the Robot Control Signal Cable "CN4" ([4] of Fig. 3.1) (Cable included)

The robot control signal cable is used for the brake ON/OFF signals for locking motor shafts; hand control signals; and others.
The robot control signal cable runs from [4] of Fig. 3.1 (CN4) to [3] of Fig. 3.2.

3.1.6 Inserting and Removing Cables

Caution

- Before inserting or removing any controller cable, be sure to turn off the "POWER" switch.
- When removing a cable, be sure to pull on the plug and not on the cord. Otherwise, you may damage the cable.

a) Circular connectors: CN3, CN4

First, push the connector on the cable side to the robot connector. Then, turn the cable side lock screw to the right to fasten in the cable. A loose plug can cause an accident or failure, so be careful and do the job right.
To remove a connector, turn the lock screw to the left and pull out the cable side connector.

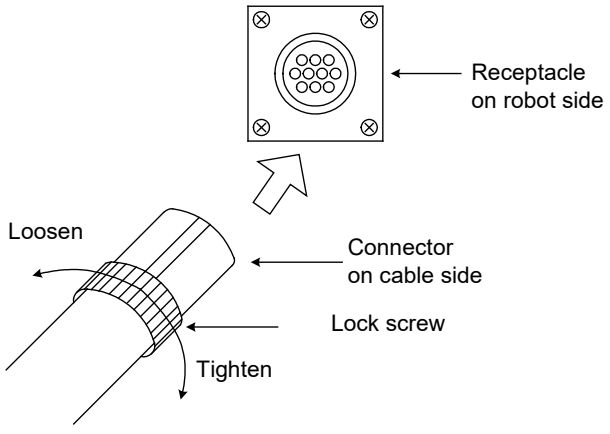


Fig. 3.3 Inserting and removing a circular connector

b) Square connectors: CN3, CN4, CN5, CN6

First, push the cable side connector to the controller connector. Then tighten the lock screws on both ends of the cable side connector with a screwdriver. A loose plug can cause a contact failure or other accident, so be careful and do the job right.
To remove the connectors, first loosen the lock screws, then pull out the cable side connector. CN4 is the quick-operated lock type connector.

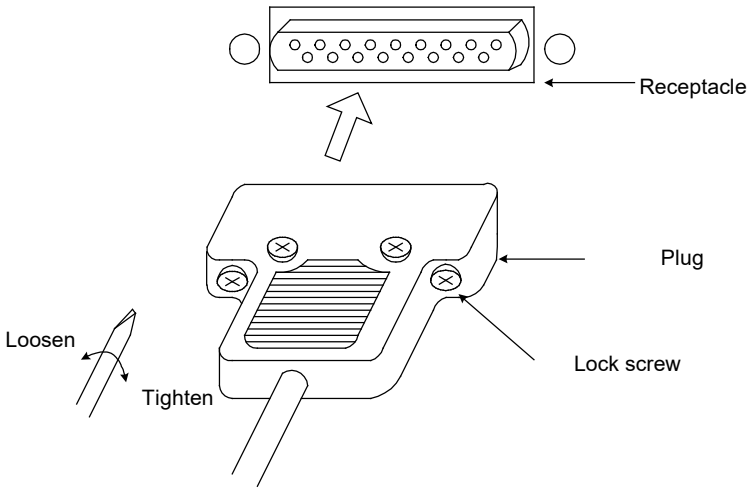
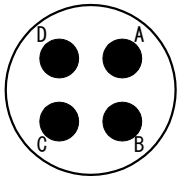


Fig. 3.4 Inserting and removing a rectangular connector

3.1.7 List of Connector Terminal Arrangement Examples

a) Power cable connector CNPW

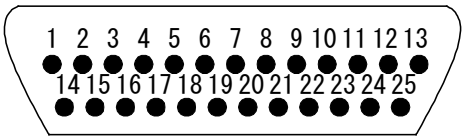


Connects to controller.
Type: JL04HV-2E22-22PE-B
Manufacturer: Japan Aviation Electronics Industry

A
B
C

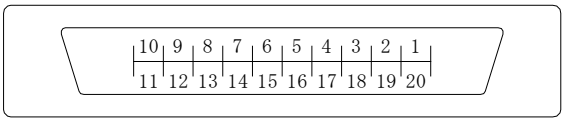
3-phase : AC180～250V
50／60Hz
Class D dedicated grounding

b) Encoder cable connector CN3



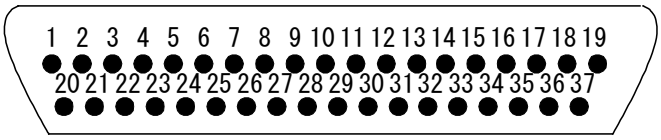
Connects to controller.
Type: FCN–777–P025–G/C
Manufacturer: Fujitsu

b) Robot control signal cable connector CN4



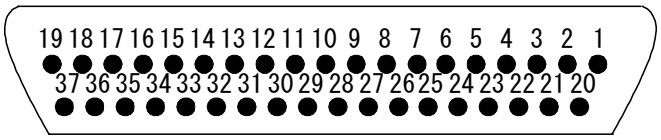
Connects to controller.
Type: PCR–E20PMC
Manufacturer: Honda Tsushin Kogyo

c) External input signal cable connector CN5



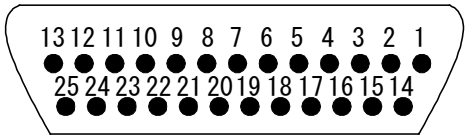
Connects to controller.
Type: 17LE–23370–27 (D3AC)
Manufacturer: Daiichi Denshi Kogyo

d) External output signal cable connector CN6



Connects to controller.
Type: 17LE–13370–27 (D3AC)
Manufacturer: Daiichi Denshi Kogyo

e) External I/O signal cable connector CN12



Connects to controller.
Type: 17LE–13250–27
Manufacturer: Daiichi Denshi Kogyo

3.2 CONTROLLER CONNECTOR SIGNALS

3.2.1 Connector Signal Connection Diagrams

Diagrams showing which signals correspond to which terminals are shown in Chapter 2 "Connecting External Cables" of the Interface Manual.

3.2.2 Jumpers for Safety Related Signals

The following external operation input signals were created for use as safety measures:

CN5-16 (Stop)
CN6-34 (Servo OFF) (Cycle operation mode)
CN5-17(Break) (Low speed command)
CN6-16,35 (Emergency stop 1B～1C)
CN6-17,36 (Emergency stop2B～2C)

The connectors provided with the controller have already been jumped for these signals. Therefore, if you wish to use these signals, you have to remove the jumpers and rewire as appropriate. Furthermore, if you plan to use the robot without external signals, always connect the connectors (provided) to the terminals of the CN5 and CN6 connectors.

Table 3.1 Connector Jumpers

CN5	CN6
16–18 17–18 (35–37) (36-19)	18–34 16–35 17-36

Caution

You will not be able to turn on the controller servo power if the Servo OFF and Emergency Stop signals have not been jumped. Also, if the cycle operation mode signal has not been jumped, the controller will operate in the cycle operation mode. If the Low Speed Command signal is not jumped, the robot will operate in low speed during automatic operation. If the Stop signal is not jumped, the robot will not be able to be operated by automatic operation.

CHAPTER 4

TOOL INTERFACE

4.1 MOUNTING THE TOOL

The tool is mounted on the tool flange. Dimensions of the tool mounting flange are shown in Fig. 4.1.
The tool is centered with either ϕ118 H7 or ϕ40 H7 mating section. The tool direction is adjusted with the ϕ8 locate pin and locked into place with six (6) M8 bolts.

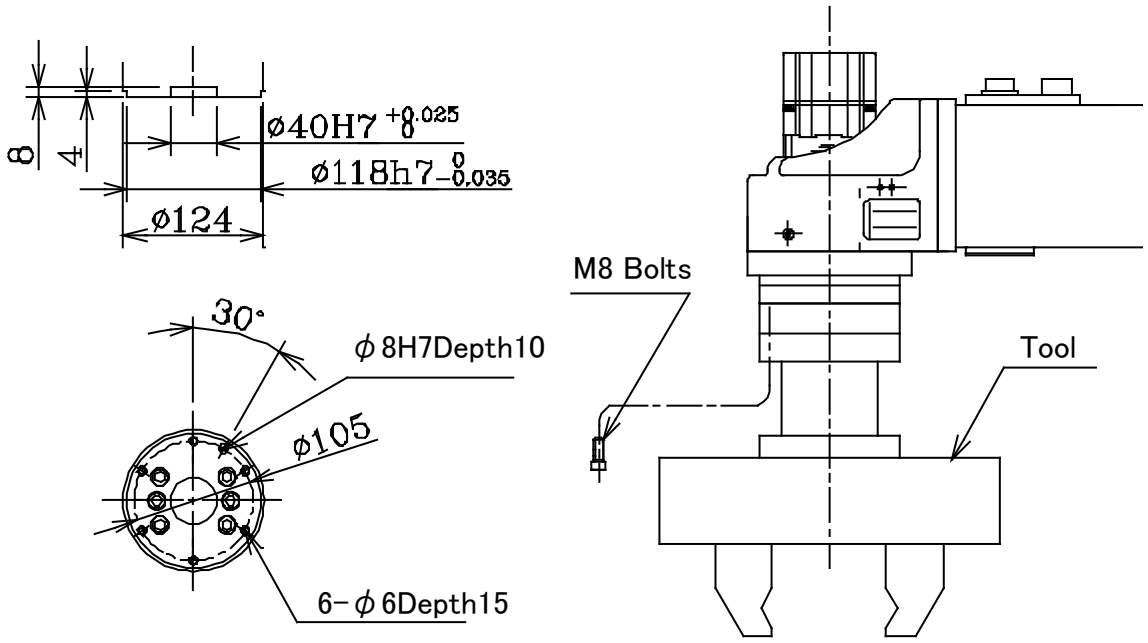


Fig. 4.1 Mounting the tool

4.2 TOOL WIRING AND PIPING

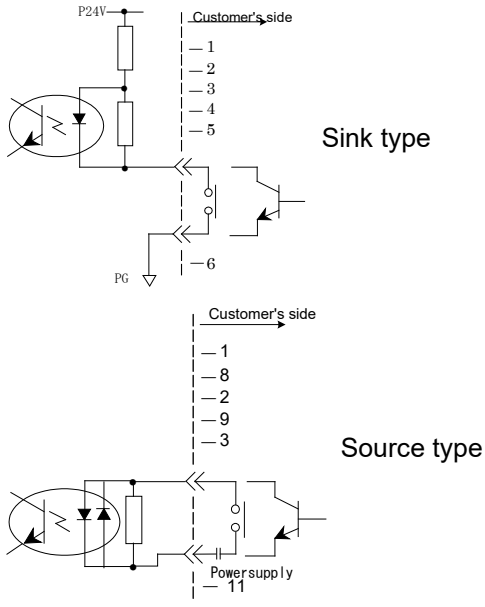
The robot is provided with wiring and air piping for attachment to the tool. These wiring and piping extend to arms and are used as follows:

4.2.1 Tool Wiring

Eight (8) input cables are provided for sensors and five (5) control signal cables are provided for solenoid valves, etc. A supply power of 24 VDC is also provided. All are connected to the controller. The wiring arrangement for these cables is shown in Fig. 4.2.

(1) Input signal Connector JOES

Signal name		Signal No.	Input circuit and example of connections
GRP1 OPN	Grip 1 open	201	
GRP1 CLS	Grip 1 closed	202	
GRP2 OPN	Grip 2 open	203	
GRP2 CLS	Grip 2 closed	204	
WORK	Work	205	
PGND	0 V		
Shield			
NC			
NC			
NC			



As input signals, no-voltage contacts or transistor open collesctor inputs are used.

No-voltage contact specifications:

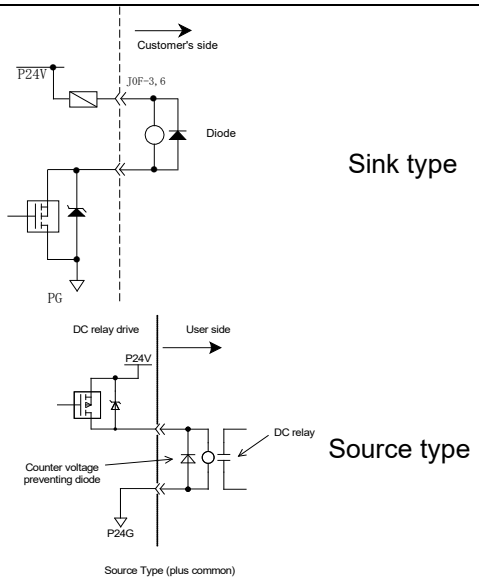
Contact rating : DC24 V, 10 mA or more (circuit current: approx. 7 mA)
Contact min. current : DC24 V, 1 mA
Contact impedance : 100 Ω or less

Transistor specifications:

Withstand voltage between collector and emitter	:	30 V or more
Current between collector and emitter (circuit current: approx. 7 mA)	:	10 mA or more
Leak current between collector and emitter	:	100 μA or less

(2) Output signal Connector JOFS

Signal name		Signal No.	Output circuit and example of connections
GRP1	Grip 1	201	
GRP1	– Grip 1	202	
GRP2	Grip 2 closed	203	
GRP2	– Grip 2	204	
NC			
P24 V	DC24 V power		
P24 V	DC24 V power		
NC			
NC			



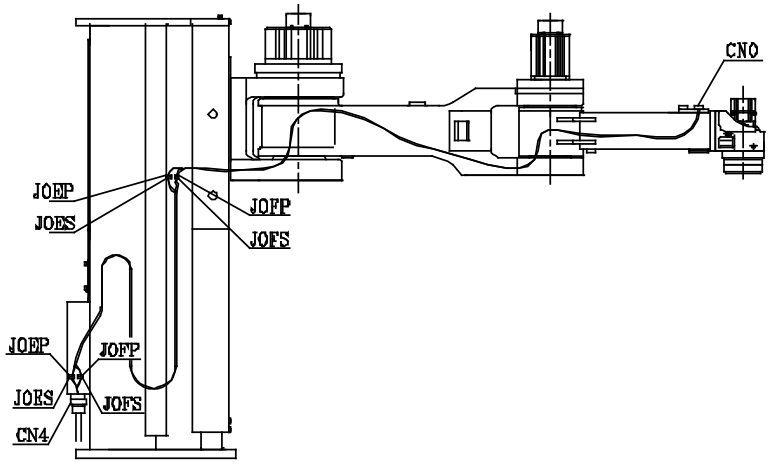
By using the 24 VDC power of the controller, a relay or solenoid valve, etc., can be driven. When the external power is used, GND of the external power should be common to the GND (PG) of SR7000 robot controller. NEVER connect +24V with P24V. Otherwise the power supply may be damaged.

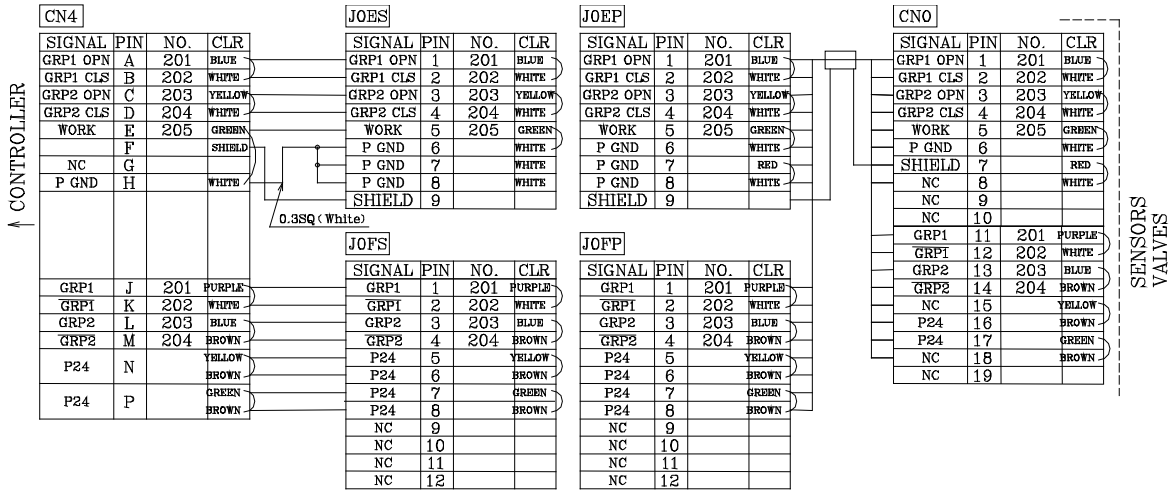
Output specifications:

Rating voltage	:	DC24 V (max. DC30 V)
Rating current	:	1 A
Leak current	:	100 μA or less

- If the 24 VDC power supplied from the robot controller is used, the total current should be 2 A or less.
- When the external power is used, the total current should also be 1 A or less.

· When a relay or solenoid valve, etc., is connected, it is necessary to use a surge killer or diode to absorb the surge voltage.





Connector types	JOES		172161–1 (Nippon Amp.)
	JOEP		172169–1 (Nippon Amp.)
	JOFS		172162–1 (Nippon Amp.)
	JOFP		172170–1 (Nippon Amp.)
	CN0	Robot side	JMR2119F (Daiichi Denshi Kogyo)
	CN0	User side	JMR2119M (Daiichi Denshi Kogyo)

Fig. 4.2 Tool wiring

The wires are connected to the connectors in the bottom cover of the arm tip. The tool cables must be drawn out through the cable outlet on the bottom side of the cover and connected to the tool.
Connector plugs are attached to the robot. The connectors are of a soldering type.

Caution

- Always use a highly flexible robot cable, and secure it below the arm with a cable clamp, etc. If a robot cable is not used, the wire could break.

In our discussion to the present, we have assumed that the tool will be controlled by a robot program from the controller. However, below we discuss what to do when controlling the tool with a separately installed type sequencer.

First, remove the pool box cover and disconnect the JOES and JOFS connectors inside. Then lead the cable from the PLC (i.e., sequencer) through the cable inlet of the cover (usually covered with grommet). Keep the current to 1 A or less per line.

4.2.2 Tool Air Piping

The robot is provided with two air lines (ϕ 12 $\times$ 2) for the tool as shown in Figs. 4.9 to 4.10. The user is responsible for obtaining and preparing the solenoid - air valves necessary to operate any pneumatic units.

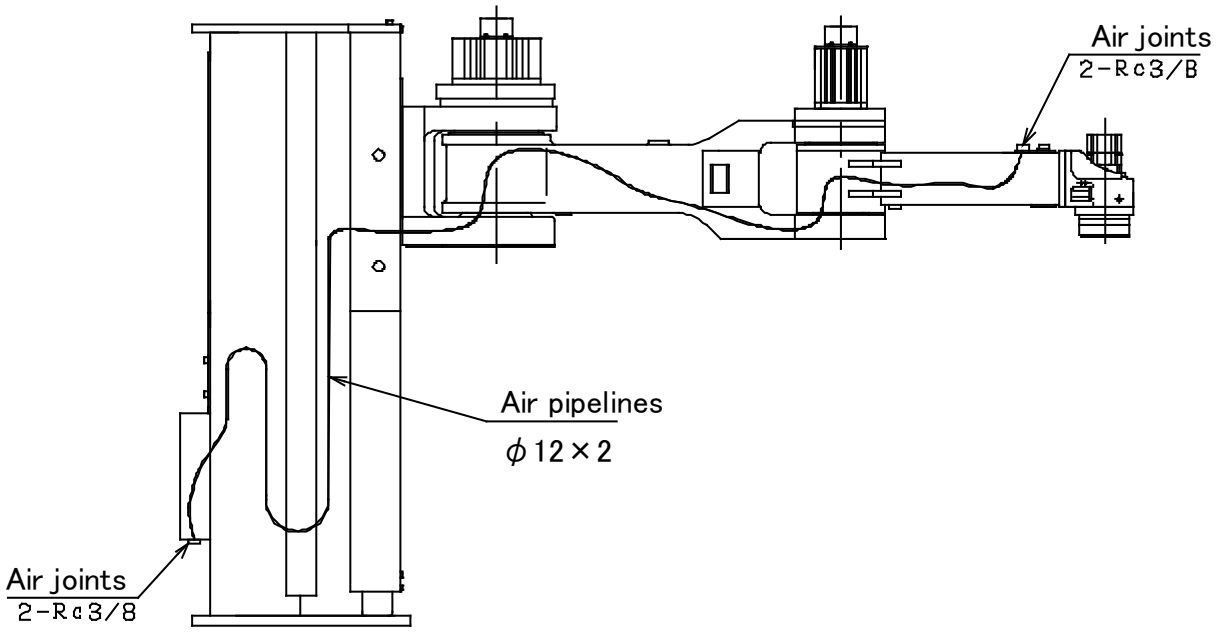


Fig. 4.3 Tool air piping

4.2.3 Mounting Solenoid Valves, etc. on Arm

When mounting solenoid valves, CCD camera, etc. on the robot arm, attach stays, utilizing M6X2 screws of wiring/piping set cover on top of the Arm 2. Dimensions of the mounting part are shown in Fig. 4.4.

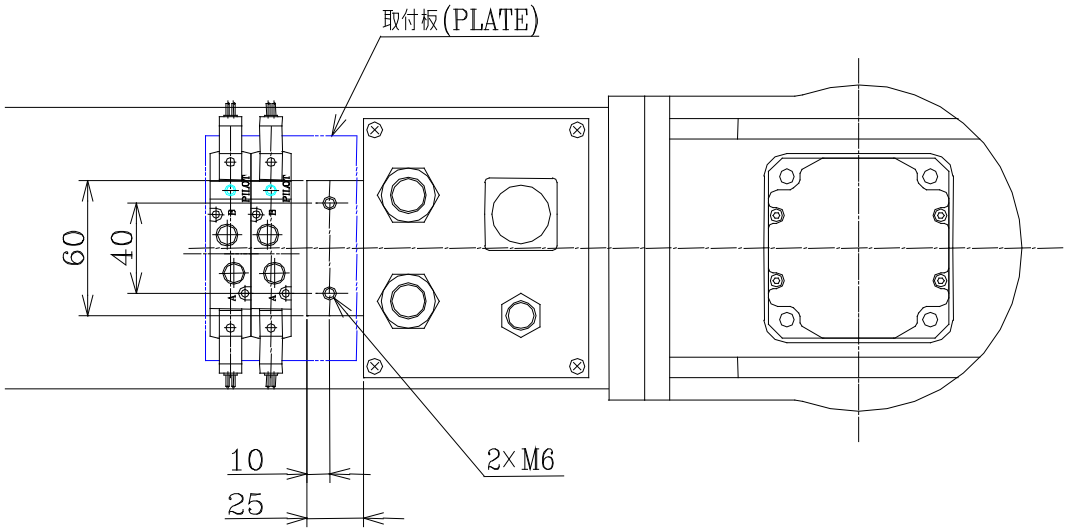


Fig. 4.4 Dimensions of mounting part (top side of Arm 2)

4.3 PERMISSIBLE LOAD CONDITIONS AND SETTING OF THE PROGRAM

This section describes the permissible load conditions of the robot and how to set the program according to the load.

4.3.1 Permissible Load Conditions

The robot load conditions are defined by the tool mass, the moment of inertia and the offset amount of the tool center of gravity from the center of the tool shaft as shown in Fig. 4.11.
The load conditions permissible for SR–1504HZ are shown in Table 4.1.

Table 4.1 Permissible load conditions

	Condition	Permissible values
SR–504HSP	Load mass	Rating 50kg (max. 70kg) Set max. acceleration for 50kg or less
	Moment of inertia	Max. 3.5 kgm^2
	Load center of gravity offset	Max. 300mm (load mass $\leq$ 30kg)

4.3.2 Load Conditions and Setting of the Program

Even if the tool load conditions are within the permissible values listed in Table 4.1, an acceleration/deceleration time and servo gain that is adequate for the load conditions must be set. If this setting is inadequate, the robot may vibrate or overshoot, and will lead to shortening the robot life.
The SR–1504 can automatically change the acceleration/deceleration time and servo gain by using the PAYLOAD command in the program according to the load conditions. The actual method for using this function is explained below.

a) PAYLOAD command
The PAYLOAD command format is written as shown below if the tool mass is M kg and the center of gravity offset is L mm:

PAYLOAD = {M, L}

M : Load mass (unit: kg)
L : Center of gravity offset (unit: mm)

The PAYLOAD command has the following functions:
- The acceleration/deceleration of each robot axis is automatically changed according to the set load conditions.
- The servo gain of each robot axis is automatically changed according to the set load conditions.

b) Program examples
Basic program examples using the PAYLOAD command is shown below. For further details, refer to the "Language Manual".

(Program example 1)
Moving the robot with load conditions of a mass of 20kg, and center of gravity offset of 200 mm.

```
PROGRAM SAMPLE
    SPEED = 100
    PAYLOAD={20, 200}
    MOVE A
    MOVE B
    STOP
END
```

(Program example 2)
When the hand mass is 10kg, center of gravity offset is 30mm, and the mass is 25 kg and center of gravity offset is 250 mm when the work is grasped.
Pick-and-place is executed with the above conditions.

```
PROGRAM SAMPLE
        PAYLOAD={10, 250}
        ACCUR COARSE
        ENABLE NOWAIT
        RESET DOUT
        MOVE TAIKI
        DOUT (1)
        WAIT DIN (1)
LOOP:
         MOVE A + POINT (0, 0,100)
         IF DIN (-1) THEN GOTO FIN
         MOVE A
         WAIT MOTION>=100
         DOUT (213)
         DELAY 1
         PAYLOAD={25, 50}
         MOVE A + POINT (0, 0,100)
         MOVE B + POINT (0, 0,100)
         MOVE B
         WAIT MOTION>=100
         DOUT (-213)
         DELAY 1
         PAYLOAD={10, 250}
         MOVE B + POINT (0, 0,100)
         GOTO LOOP
FIN:
         MOVE TAIKI
         DOUT (1)
         STOP
END
```

c) Setting the PAYLOAD command
In the default state, or when the PAYLOAD command is not used, the acceleration/deceleration will be set to 100% and the servo gain will be set to the value for the minimum load. (See Section 4.3.3.)

Caution

Thus, use the PAYLOAD command in the following cases:
・ When load mass exceeds the rated load
・ When there is a load center of gravity offset
If the PAYLOAD command is not used, the robot will vibrate or overshoot, and will shorten the life of the mechanisms.

Caution

The load inertia must be within the tolerable values given in Table 4.1. Even if there is no load center of gravity offset, if the inertia is large, the robot may vibrate. In this case, calculate a hypothetical offset L mm with the following equation using the inertia J kg・m^2 and mass Mkg. Then, designate the following:

$L = \sqrt{J \times 10^6 / M}$

PAYLOAD = {M, L}

Caution

When guiding manually, the robot may vibrate if the load if the load mass or center of gravity offset is large. This is caused because the servo gain is not appropriate. In this case, perform the following while setting the load conditions in the test run mode.

DO PAYLOAD = {30, 300}
EXECUTE
The servo gain will be change to that which meets the load conditions.

4.3.3 Setting Robot Acceleration/Deceleration Time for Load Conditions

a) When offset is absent
The robot's acceleration/deceleration time will be automatically changed according to the load conditions when the PAYLOAD command is used.
The acceleration/deceleration time will change as shown in Fig. 4.6 according to the load mass. The vertical line shows the acceleration during acceleration/deceleration. For example, if the load mass is 70 kg, the maximum acceleration will be 75 %.

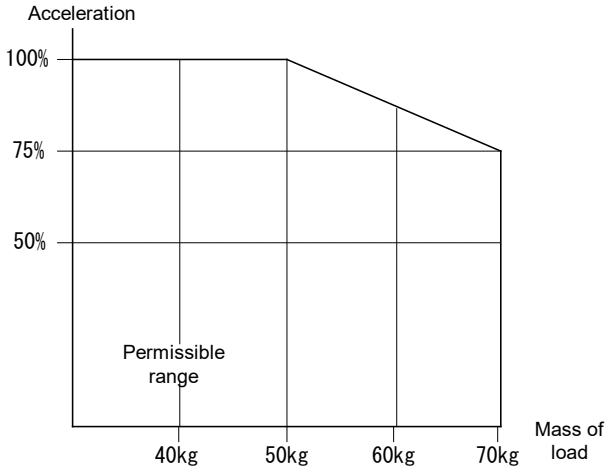


Fig. 4.6 Setting of acceleration to load mass

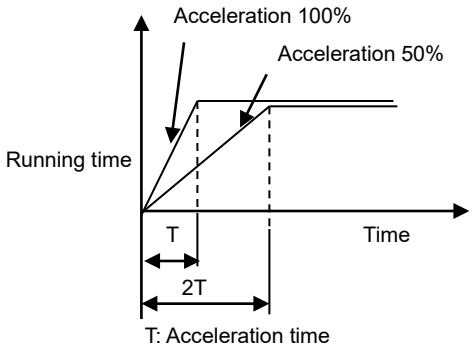


Fig. 4.7 Acceleration changes

b) When offset is present
When an offset of load is present, acceleration is limited by the offset value. The maximum offset of load should be 300 mm (mass of load 30 kg).
The figure below shows the rate of reduction of acceleration according to the offset value.

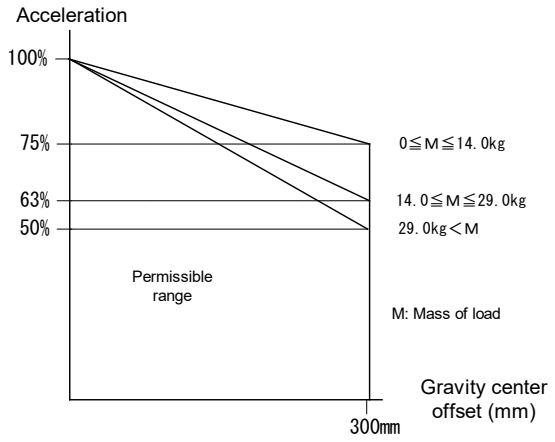


Fig. 4.8 Permissible load conditions (with offset)

APPENDIX A Calculating Control Power Capacity

The power capacity of the controller can be calculated in the following manner. The power connected to the controller should exceed the capacity obtained from the following equation:

(Power capacity) = (Power capacity of control section) + (Power capacity of servo driver)

Robot model	Power capacity of control section (VA)	Power capacity of servo driver (VA)	Power capacity of controller (kVA)
SR–1504HZ	450	7550	8

Note) With respect to the power capacity required, it is assumed that all the axes are continuously operated at 100% load. If the power capacity is insufficient when the robot is operated, troubles such as a drop in power voltage, a malfunction, or resetting of the control power may take place. Thus, the robot may not be correctly operated. In addition, a current three times the rating current may be required at start of the robot. Thus, it is necessary to select a power supply in which the peak current of the power is three times the rating current for 100 ms.

APPENDIX B Restoring Zero Position Data (Restoration by Visual Check of Zero-Point Marks)

Batteries which back up the positional information of the position detectors (i.e., encoders) are contained in the robot base unit. Unless the robot is used for a long period of time, the battery voltage drops and the home position data will be lost. Also, if the voltage of the battery which backs up the internal memory of the X8HC printed board drops, programs, parameters and absolute coordinate data set by the user will be lost. To avoid this, replace the battery on a regular basis. The yardstick of each replacement period is shown below.

- Batteries for position detectors ⋯ Two (2) and a half (2.5) years after delivery (when the operating time per day is eight (8) hours).
- Battery for X8HC printed board SRAM ⋯ Five (5) years after delivery

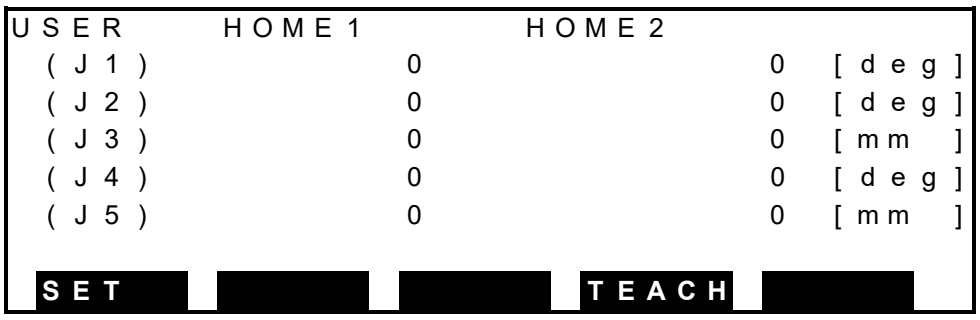
If the home position data has been destroyed due to battery voltage drop, etc., observe the following procedures to restore it.

a) How to restore the home positions

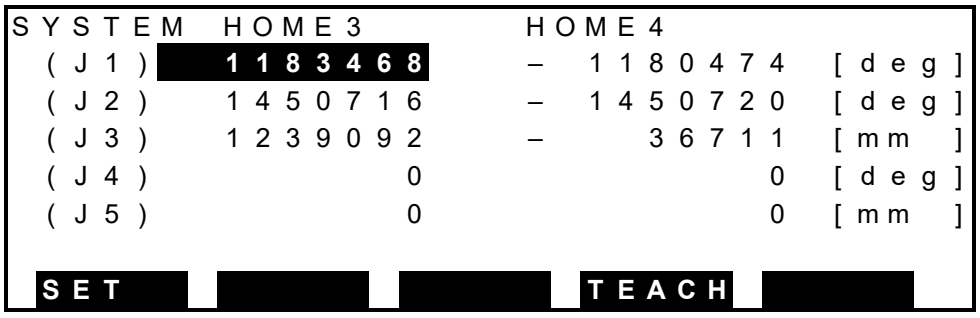
(1) Descriptions of HOME function

This function is useful to restore the original home position data by memorizing each axis coordinate data beforehand, if the machine home position data has been destroyed.
The set points can be assigned to HOME1 ~ 4. The plus (+) side mechanical stopper positions of axes 1 ~ 3 are factory-set in HOME3, and the minus (–) side mechanical stopper positions of axes 1 ~ 3 are factory-set in HOME4. The axis 4 is set at the 0° position according to the match-mark. In HOME1 and HOME2, the user can specify any desired positions. We recommend to set there the positions securing the robot end to prepare for the system construction.

[Page 1]



[Page 2]



Note: The above are the screens for restoring the machine home positions. NEVER operate them at other than the restoration.

(2) Operating procedures

I. How to set the data in HOME1 and HOME2:

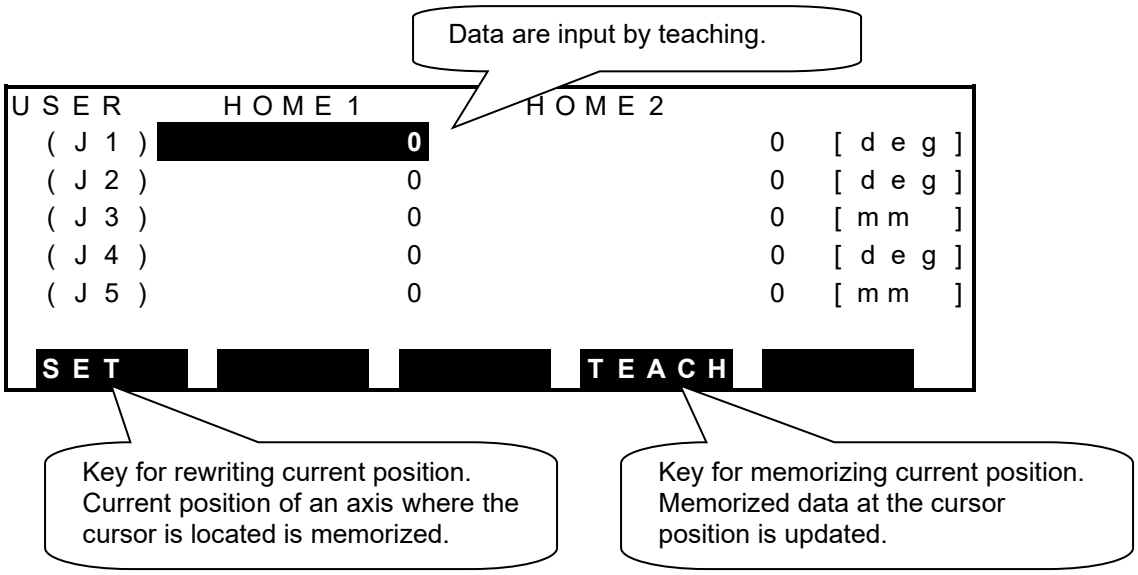
This paragraph describes the data setting in HOME1. (The same procedures are applied to HOME2.)

[1] Guide the robot to secure the tool shaft at a desired position.

[2] Turn off the servo power.

[3] Press the [UTILITY] key through the teach pendant.

[4] Press the NEXT key until [REORG] appears, then press the [REORG] key.



[5] When the above screen appears, memorize the position where the robot was secured in the [TEACH] mode, by moving the cursor to each axis.

II. How to restore the home position data by HOME1 (or HOME2):

If the machine home position coordinate data has been destroyed, the machine home position can be restored by means of the HOME1 function.

[1] Guide the robot to the position set by HOME1 and secure the robot there.

[2] Turn off the servo power.

[3] Press the [UTILITY] key through the teach pendant.

[4] Press the NEXT key until [REORG] appears, then press the [REORG] key.

[5] Move the cursor to each axis in the HOME1 mode of the REORG screen. Press the [SET] key and write the position where the robot was secured to the coordinate value set in HOME1. Thus, the machine home position can be restored. When you are going to set the data to axis 3 at this time, be sure to set axis 4 first.

III. How to restore the position data by HOME3 (or HOME4):

The plus (+) side mechanical stopper positions of axes 1 ~ 3 are factory-set in HOME3, and the minus (–) side mechanical stopper positions of axes 1 ~ 3 are factory-set in HOME4. The axis 4 is set at the 0° position according to the match-mark.

The machine home position can be restored, using these data.

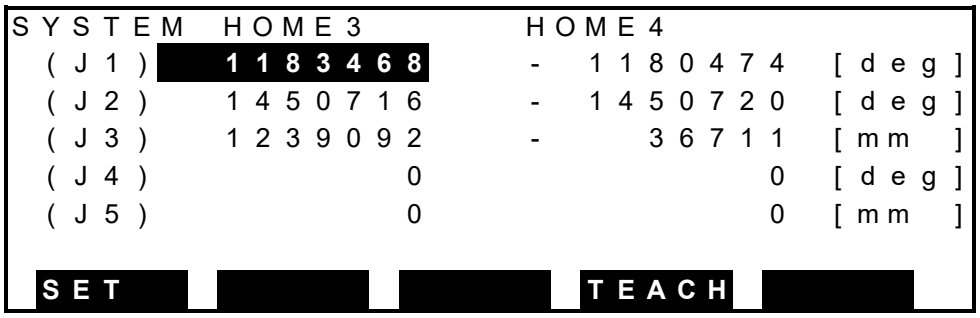
This paragraph deals with the restoration procedures, using HOME3.

[1] Turn off the servo power.

[2] Press the [UTILITY] key through the teach pendant.

[3] Press the NEXT key until [REORG] appears, then press the [REORG] key.

[4] Press the NEXT key to display Page 2.



[5] Contact the axis 1 of the robot to the plus (+) side mechanical stopper by hand.

[6] Move the cursor to (J1) of HOME3, then press the [SET] key. Now, the machine home point of axis 1 is restored.

[7] Likewise, contact the axis 2 of the robot to the plus (+) side mechanical stopper by hand.

[8] Move the cursor to (J2) of HOME3, then press the [SET] key. Now, the machine home point of axis 2 is restored.

[9] Press the [ESC] key to escape from the REORG screen once.

[10] Turn on the servo power, and set the axes 3 and 4 in the servo-free state.

[11] Set the axis 4 to the match-mark.

[12] Contact the axis 3 of the robot to the plus (+) side mechanical stopper by hand.

[13] Turn off the servo power.

[14] Repeat Steps [2] to [4] inclusive to call the HOME screen again.

[15] Move the cursor to (J4) of HOME3, then press the [SET] key. Now, the machine home point of axis 4 is restored.

[16] Move the cursor to (J3) of HOME3, then press the [SET] key. Now, the machine home point of axis 3 is restored.

[17] Press the [ESC] key to escape from the HOME screen. Now, the operation has finished.

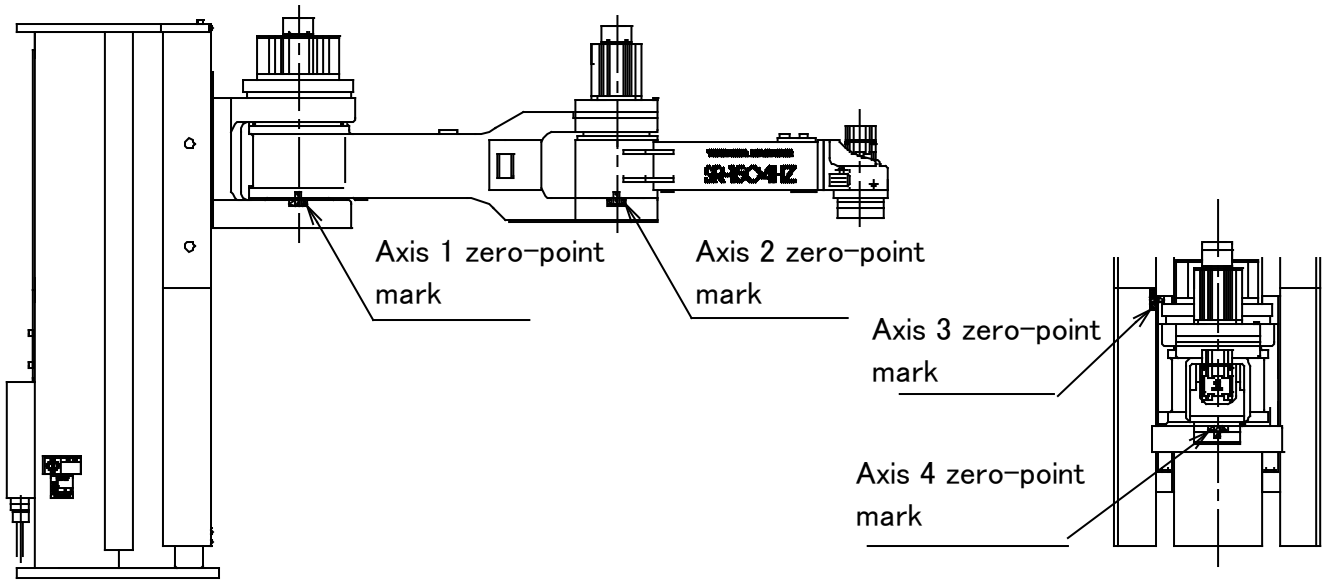


Fig. A.1 Positions of zero-point marks

APPENDIX C Restoring Zero Position Data (Restoration by Zero Point Setting Tool)

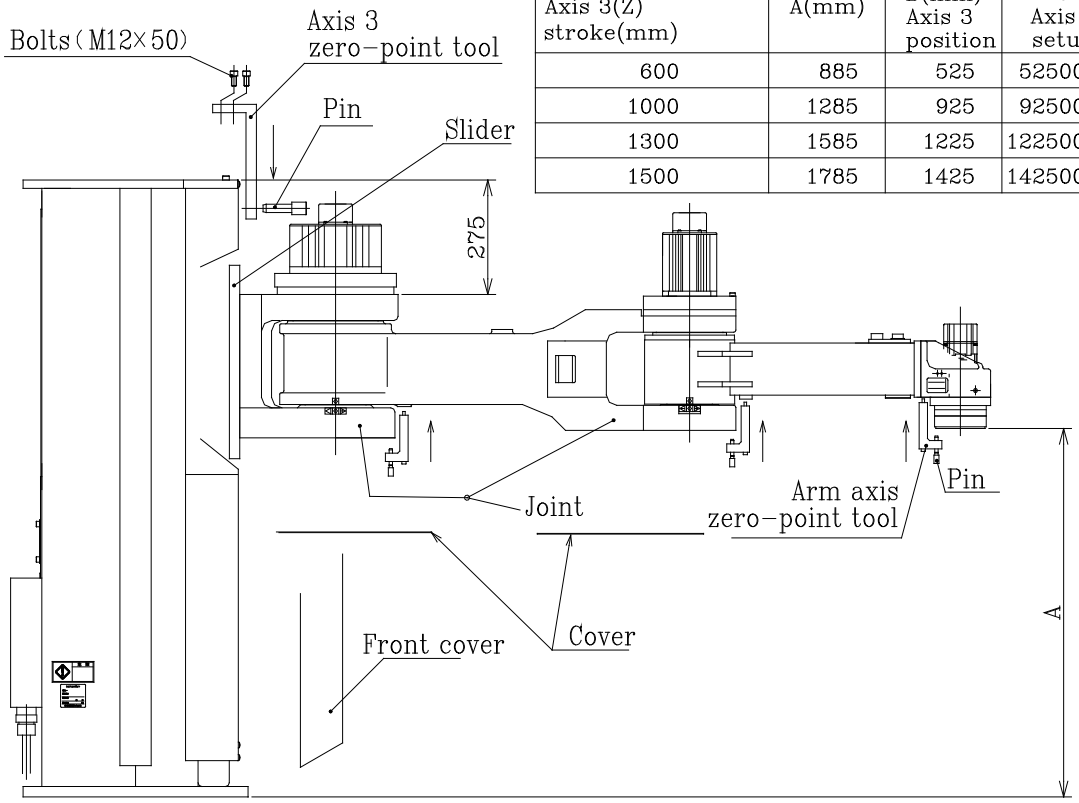
This section describes the procedures for restoring the zero position data, using the zero point setting tool (option).

a) Turn on the power of the controller, and align the zero-point marks of a desired axis in the manual mode.

a–1) When restoring Axes 1, 2 and 4:
If an axis to be restored is the Axis 1 or 2, remove the cover on the bottom side of the joint. If it is the Axis 4, remove the hand. Insert the zero-point setting tool into the two (2) holes (ø8) on the arm or on the bottom side of the joint and secure with attached bolts (M6×120).
Insert the locate pin into the hole (ø8) of the joint or tool flange. At this time, if the joint or tool flange position is shifted, manually guide it to the aligned position. This position is the zero position of selected axis.

a–2) When restoring Axis 3:
Remove the robot front cover. Guide the axis, while observing with a scale, etc., so that the length between the joint 1 top surface and robot column top surface is about 275 mm. Secure the Axis 3 setting tool to the column top surface with two (2) bolts (M12×50).
Insert the locate pin into the long hole of the slider. At this time, if the slider position is shifted, manually guide it to the aligned position. This position is "B value of fig.A.2" mm.

b) Execute the steps in Appendix B–b) above. When setting the zero position of Axis 3, however, enter "C value of fig.A.2".



Axis 3(Z) stroke(mm)	A(mm)	B(mm) Axis 3 position	C Axis 3 setup
600	885	525	5250000
1000	1285	925	9250000
1300	1585	1225	12250000
1500	1785	1425	14250000

Fig. A.2 Use of zero point setting tool